Solarthermal controller wiring (Waveshare ESP32-S3-POE-ETH-8DI-8RO)

Wire colour = conductor colour where documented. Pale tint = white-striped wire (Rob: in a pair, the wire with white is the ground). Grey = colour not documented, marked "?". Dashed = planned, not wired yet.

Spa (outdoors)

SPA1 — Spa controller box, Blue CAT5

CALL+	Blue: 11.2 V calling	SPA_CALL+
CALL−	Blue/White: 0 V path when idle	DGND
ENABLE+	Orange: to heater relay board	SPA_ENABLE+
ENABLE−	Orange/White	GND

T3 — Spa temp, Grey CAT5, on pipe (relative only)

+5V	Brown	+5V
GND	Brown/White	GND
OUT	Orange/White	TEMP_SPA

V3 — Spa valve: power opens, loss of power closes

SPA_VALVE+	OPEN+	
GND	RET−	

Grey CAT5 Blue, Blue/White (old motor pair). Check with a meter before wiring R8.

Box power

PS1 — 12 V supply

AC_L ?	L	+12V → +12V
AC_N ?	N	GND → GND

PS2 — DIANN 12 V to 5 V, 3 A

+12V	IN+	+5V → +5V
GND	IN−	GND → GND

Temperature inputs (0–5 V, TMP36-type: (mV−500)/10 C)

T1 — Tank

+5V	?	+5V
GND	?	GND
OUT	?	TEMP_TANK

T2 — Left panel

+5V	?	+5V
GND	?	GND
OUT	?	TEMP_LEFT

T4 — Right panel

+5V	?	+5V
GND	?	GND
OUT	?	TEMP_RIGHT

U2 — Modbus RTU Analog Input 8CH, addr 1, 9600 8N1

TEMP_TANK	? colour	AI1+
GND		AI1−
TEMP_LEFT	? colour	AI2+
GND		AI2−
TEMP_SPA	Orange/White	AI3+
GND		AI3−
TEMP_RIGHT	? colour	AI4+
GND		AI4−
AI1+	7–36V	+12V
AI1−	GND	GND
AI2+	A+	RS485_A
AI2−	B−	RS485_B

U1

8DI-8RO, static 192.168.1.9

VIN unused? PoE powers U1	VIN+	R1_COM: +12 V → +12V
	VIN−	R1_NO: ? solar pump SSR + → SSR1_IN+
RS485_A ? colour	A+	R1_NC
RS485_B ? colour	B−	R2_COM: +12 V → +12V
	PE	R2_NO: ? recirc pump SSR + → SSR2_IN+
		R2_NC
	COM: leave empty, note 6	R3_COM: +12 V → +12V
DGND isolated, not box GND	DGND	R3_NO: ? Takagi SSR + → SSR3_IN+
SPA_CALL+ Blue → D1 A	DI1 Blue, band toward DI1 → D1 K	R3_NC
V1_CW Green? see note 3	DI2	R4_COM: existing red → +12V
V1_CCW Brown? see note 3	DI3	R4_NO: Orange (Blue CAT5) → SPA_ENABLE+
V2_CW ? note 3	DI4	R4_NC
V2_CCW ? note 3	DI5	R5_COM: +12 V → +12V
	DI6	R5_NO: ? to data closet → POOL_REQ+
	DI7	R5_NC
	DI8	R6_COM: +12 V → +12V
		R6_NO: ? to DPDT pole A common → ROOF_POWER
PoE switch, LAN	POE	R6_NC
		R7_COM: +12 V → +12V
		R7_NO: ? to DPDT coil + → DPDT_COIL+
		R7_NC
		R8_COM: +12 V → +12V
		R8_NO: Blue (Grey CAT5)? → SPA_VALVE+
		R8_NC

D1 — silicon diode

Notes

1. U1 relay wiring is from the proposed table and Rob's reports; "?" colours aren't documented.
2. R4: "existing red" and orange come from the proposed table; check on the wire.
3. DI2–DI5: Brown CAT5 labels give Green/Brown/Green-White for one valve. Valve 2's colours are unknown.
4. Valve internals and motor-limit wiring are deliberately not drawn.
5. Diode D1 stops idle current through the spa's low-resistance output.
6. Commons go on DGND. With them on COM, DI1–DI5 all read inactive.
7. R7 drives DPDT K4: energized = pool, released = tank. R6 feeds motor power, on for 60 s per move. D2 band to coil +.

Pool controller (over 100 ft, spliced in data closet)

U3 — poolpumpcontrol 8RO board

POOL_REQ+ ?	DI2	
GND ?	DGND	

Mains side (SSRs switch the load; relays carry only SSR control)

K1 — Solar pump SSR, 3–32 VDC in

SSR1_IN+ ?	IN+	AC1 ? mains → AC_L
GND ?	IN−	AC2 → AC_N

M1 — Solar pump: HOT, NEUT

K2 — Recirc pump SSR, 3–32 VDC in

SSR2_IN+ ?	IN+	AC1 ? mains → AC_L
GND ?	IN−	AC2 → AC_N

M2 — Recirc pump: HOT, NEUT

K3 — Takagi heater SSR, 3–32 VDC in

SSR3_IN+ ?	IN+	AC1 ? mains → AC_L
GND ?	IN−	AC2 → AC_N

M3 — Takagi heater: HOT, NEUT

Roof (both valves on one set of motor wires; each has separate motor-limit and position switches)

V1 — Roof 3-way valve 1, internal wiring not drawn

ROOF_B Blue: valve blue wire	MOTOR_BLU	CW_TANK Green: valve blue → V1_CW
ROOF_BW Blue/White: valve white	MOTOR_BLU/WHT	CCW_POOL Brown: valve yellow → V1_CCW
ROOF_O Orange: valve red	MOTOR_ORG	POS_COMMON Green/White: valve orange → DGND
ROOF_OW Orange/White: valve black	MOTOR_ORG/WHT	

V2 — Roof 3-way valve 2, internal wiring not drawn

ROOF_B Blue: valve blue wire	MOTOR_BLU	CW_TANK ?: valve blue → V2_CW
ROOF_BW Blue/White: valve white	MOTOR_BLU/WHT	CCW_POOL ?: valve yellow → V2_CCW
ROOF_O Orange: valve red	MOTOR_ORG	POS_COMMON ?: valve orange → DGND
ROOF_OW Orange/White: valve black	MOTOR_ORG/WHT	

Tank = 1010 and pool = 0101 on DI2–DI5. Solar pump needs 1010; pool request needs 0101.
A commanded move not confirmed in 15 s sends an URGENT alert.

Roof valve DPDT relay K4, 12 V coil, in the controller box (planned)

K4 — DPDT relay, 12 V coil

DPDT_COIL+ ? from R7 NO	COIL+	A_NO: Blue, Brown CAT5: pool + → ROOF_B
GND box ground	COIL−	A_NC: Orange, Brown CAT5: tank + → ROOF_O
ROOF_POWER ? from R6 NO	A_COM	B_NO: Blue/White: pool − → ROOF_BW
GND box ground	B_COM	B_NC: Orange/White: tank − → ROOF_OW GND

D2 — flyback, across K4 coil: A → DPDT_COIL+, K

Drawn from the wiki, old sketch comments, cable labels and Rob's reports

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